

## **Element C: Presentation and Justification of Solution Design Requirements**

### **Design Requirement #1: Safety (HIGH Priority)**

The design must be safe to use. It cannot contain multiple pieces, fragile unsecured objects, or be able to injure anyone. If the design presents some sort of danger than it cannot be used. The design solution is made from a single piece, and the voting objects are pom-poms, and thus neither the voting objects nor the challenge station itself can cause harm.

The primary stakeholder stated that safety is a high priority and is one of the necessary requirements for the design solution.

### **Design Requirement #2: Durability (HIGH Priority)**

The design has to be durable enough to survive near constant use every day. If the design is not durable enough it will quickly break and become unusable, thus durability is a very important consideration for the design. The design solution is made from a single piece and built to be sturdy under constant proper use.

There was no feedback.

### **Design Requirement #3: Dimensions (HIGH Priority)**

The design has to fit on the roughly 11 by 18-inch desk. If the design is too large it cannot be selected. The design solution my team came up with is designed to be as compact as possible.

Specific dimensions were supplied by the primary stakeholder that the design solution had to follow.

### **Design Requirement #4: Fun level (HIGH Priority)**

The design must be fun and enjoyable to use. The entire purpose of the design is to provide an enjoyably activity to its users. The solution is designed to be physically interactive, by forcing the user to perform active motion in a manner similar in concept to throwing a basketball into a hoop. This makes the design inherently competitive and fun.

There was no feedback.

### **Design Requirement #5: Cost (LOW Priority)**

The cost of the design must be quite low. No matter how well it fulfills other design requirements, a design cannot be constructed if it exceeds its budget. The intended material to construct the design solution are 3D printed substances, thus making the cost of this design extremely low. In addition, there are no moving parts or complex components.

The primary stakeholder stated that if the design solution was too expensive it would not be economically viable to use it.

### **Design Requirement #6: Simplistic Construction (LOW Priority)**

The ease of use of the design must be high. The intended users are middle schoolers, and they must be able to operate it without exterior assistance. The design solution is a single solid object, with no moving parts or any components that would require instruction. The solution was also designed to make its intended use obvious.

The primary stakeholder stated that the users of the design would be middle schoolers, and that the design must therefore be easy to use.

## Problem definition

A local middle school librarian Mrs. Hybarger wants a this or that game to keep kids engaged while at the checkout desk. And to vote on things like which problem they want solved or what activities they'd rather do.

## Problem justification

When Middle Schoolers are forced to go to the library they can often be disengaged or bored.

### Built design photos



### Feedback

| Feedback                                                                                                                                                               | Response                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Feedback from Mrs. Hybarger: The design is too small and the voting boxes are too close together. The design is too small and the voting boxes are too close together. | Response: The design is too small and the voting boxes are too close together. The design is too small and the voting boxes are too close together. |
| Feedback from Mrs. Hybarger: The design is too small and the voting boxes are too close together. The design is too small and the voting boxes are too close together. | Response: The design is too small and the voting boxes are too close together. The design is too small and the voting boxes are too close together. |

### Testing plan

| Test | Test Description                                                                                      | Test Results | Test Conclusion                                                                                       |
|------|-------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------|
| 1st  | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. | Pass         | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. |
| 2nd  | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. | Pass         | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. |
| 3rd  | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. | Pass         | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. |
| 4th  | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. | Pass         | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. |
| 5th  | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. | Pass         | The prototype must hold the voting boxes in place. The prototype must hold the voting boxes in place. |

### Materials List

| Material name | Description | Where material was purchased | Cost per unit | # units | Total cost per item |
|---------------|-------------|------------------------------|---------------|---------|---------------------|
| cardboard     | cardboard   | N/A                          | 2             | 23      | 0                   |
| wood          | wood        | craft store                  | 1             | 1       | 1                   |
| glue          | glue        | craft store                  | 1             | 1       | 1                   |
| wood          | wood        | craft store                  | 1             | 1       | 1                   |

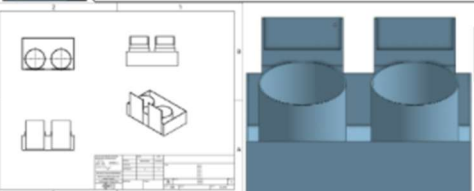
### Initial Ideas



### Decision matrix

|            | Design 1 (cardboard) | Design 2 (wood) | Design 3 (wood) | Design 4 (wood) |
|------------|----------------------|-----------------|-----------------|-----------------|
| Size       | 1.0                  | 1.0             | 1.0             | 1.0             |
| Stability  | 1.0                  | 1.0             | 1.0             | 1.0             |
| Weight     | 1.0                  | 1.0             | 1.0             | 1.0             |
| Cost       | 1.0                  | 1.0             | 1.0             | 1.0             |
| Complexity | 1.0                  | 1.0             | 1.0             | 1.0             |
| Score      | 1.0                  | 1.0             | 1.0             | 1.0             |

### Prototype/Prototype Build



The original idea of the design is you have two voting boxes side by side that are designed similar to a basket ball hoop and the users will throw pompoms into the hoops to vote.

### New Iterations

After careful review of all the feedback given from stakeholders and subject experts we made 3 main changes.

To hold the note cards we added on the backboard we designed it so that a notecard can be slotted into place.

To hold the pom pots before use we have extended the base to the max of 11x18 and then added sidewalls so that any unused pom pom or pom pom that was thrown and missed did not go on the ground.

We also moved each "hoop" farther apart so that accidental wrong votes will not happen and have them not diagonal not the board so they sit better on the table.

### Videos

#### Design demonstration



#### Demonstration.mov

#### Testing video



#### Test.mov